

MAE 2250: Mechanical Synthesis

ODP 7

Adapticon Solutions

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Display of 8 Renderings:

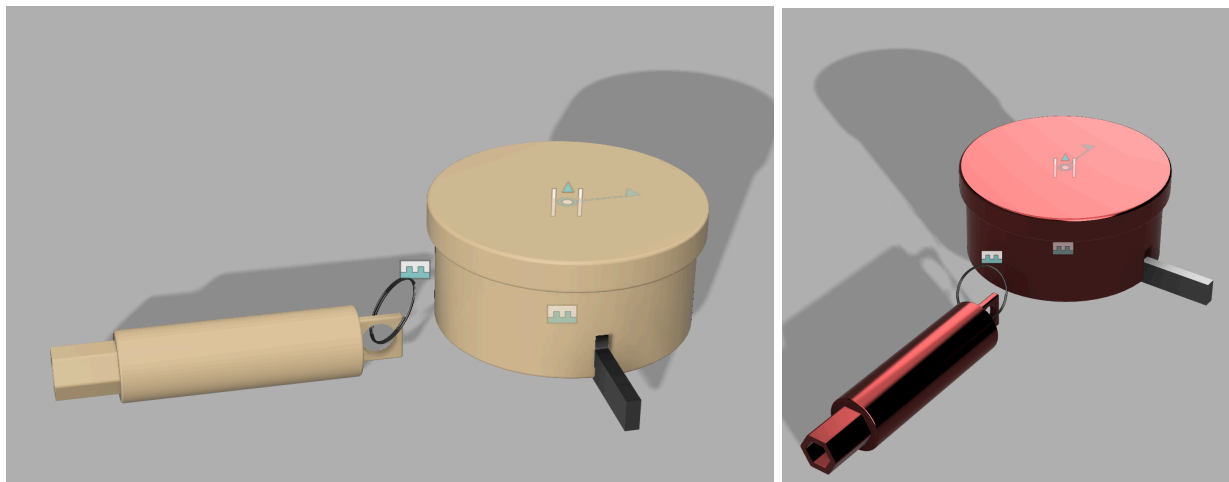
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Design for Manufacturing

The Adapt-a-bit was designed with the purpose of serving as a small and lightweight storage container that helps individuals perform on the go construction tasks. With this specific function in mind, we found it necessary to manufacture the bulk of our components via 3-D printing while outsourcing the minor parts. This allowed us to produce multiple prototypes for testing and assessment purposes while also staying comfortably within budget. The simple nature of the design process led to a fairly simple manufacturing process which resulted in aesthetics and ergonomic features being sacrificed.

Using sleek metals like aluminum, stainless steel, and tin would improve the appearance of the Adapt-a-bit multi-functional keychain and it would still fit the lightweight criteria. In addition, these types of metals generally have soft surfaces which would be beneficial for comfortable usage. Furthermore, using metals would improve the durability of the Adapt-a-bit when the attachment and hex bit are used. For the case exterior and the case top can be manufactured using a cnc machine to round out metal blocks while also creating the threads for case exterior and case top. A waterjet machine can be used to detail the specific compartments of the case interior.

Another option is to use plastic or rubber due to the ease of manufacturing the product in mass due to the low cost. Both materials are very lightweight and the smooth surface can attract users who are looking for enhanced comfort and minimizing strain on the hand when performing construction tasks. In addition, both materials are safer to use and there is a decreased risk of getting injured while using the tool. Plastic moulding can be performed to manufacture all the components if the moulding containers are designed based on the measurements of each component. 3D printing could also remain the main source of manufacturing the case exterior, case interior, case top, and accessible due to the ease of it and prior success with it



Design For Assembly:

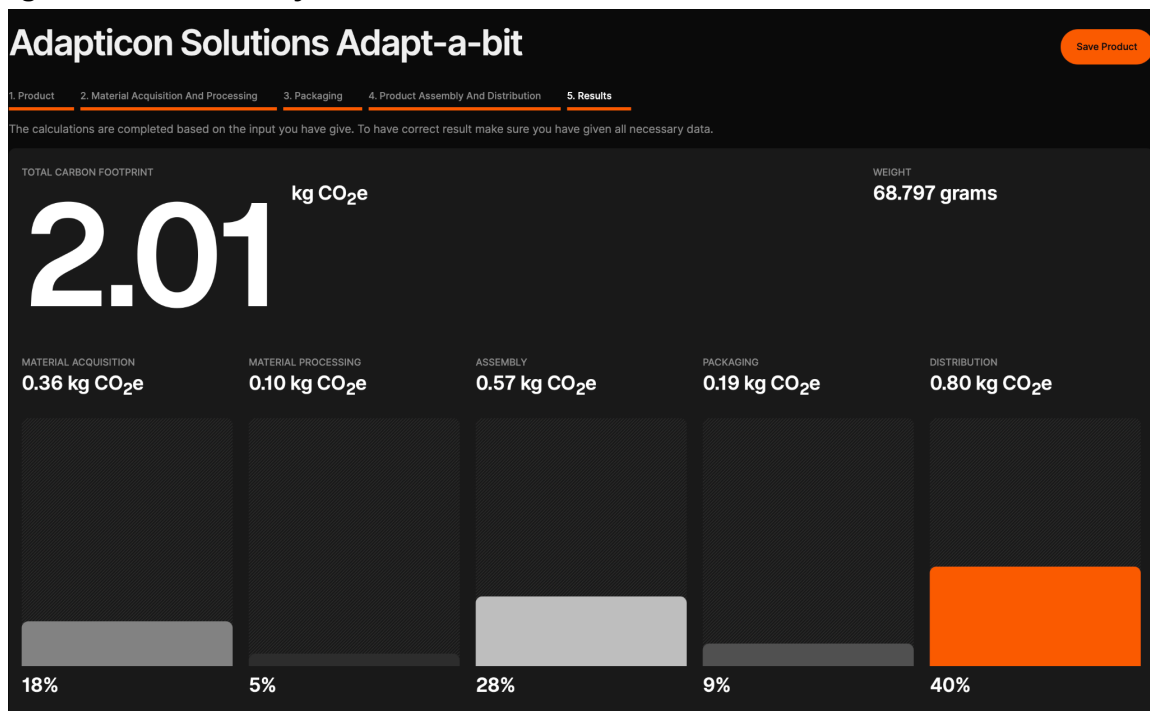
Our final design has 4 manufactured components, the case exterior, case interior, case top, and accessible attachment, as well as two bought parts, the tape measurer and key chain ring. To assemble our design you coil the measuring tape on the bottom of the interior and place the interior part inside the exterior, making sure the measuring tape is placed into its exit point through the exterior design so it can function properly. Then twist on the screw on top to finish the case assembly. Lastly, you connect the case with the keychain ring on the applicable extrude and then connect the attachment piece with the ring as well.

The key advantages of our assembly is that it can be manually assembled with any tools or machinery. Additionally, assembling the design is fairly simple and would be easy for a consumer to take apart if needed for maintenance. The frustrating part of assembly comes from the measuring tape mechanism. It can be really difficult to coil the measuring tape while its slotted in its exit point. The interior isn't constrained in the direction orthogonal to the base and top of case so if the interior is moved up from the case, either by coincidence or when rotating the interior for the measuring tape, it can uncoil and mess up the measuring tape and you would then have to recoil it.

To improve the assembly process for the design, one change is that we can have the exterior and interior of the case be one part, removing one tedious step in assembly. By doing this, we now need a new entry point to attach the measuring tape. The second change would be making it so the base on the design can be taken on and off. This would allow easier access to attach the measuring tape to the bottom of the case. The base cap is a clip on attachment rather than the screw on attachment so that you don't mix up which cap goes where. Because we made these changes, the spring can't be a manual system now. The third change would be to add a spiral spring to accompany the measuring tape. The spiral spring is connected with the measuring tape and allows the measuring tape to retract when pulled forward. This change is necessary so that design assembly is improved whilst still maintaining the design's mechanism. This change also reduces the amount of filament needed at the bottom of the previous case interior as it's supported differently and requires less material there.



Design For Sustainability:



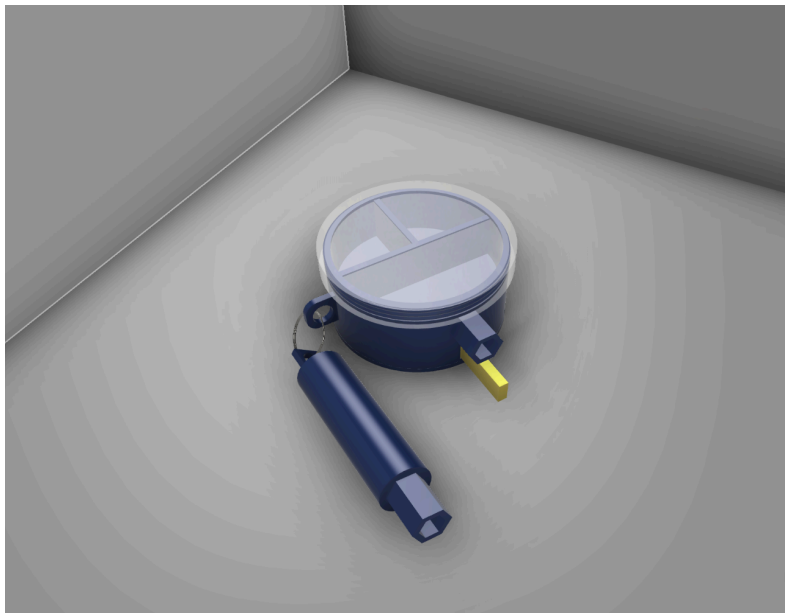
The current design of the Adapt-a-bit multifunctional keychain reflects a strong intent toward utility and portability, but sustainability considerations reveal several areas for improvement. At 68.8 grams and a total carbon footprint of 2.01 kg CO₂e, the most significant contributor to its environmental impact is distribution, which accounts for 40% of total emissions. This is followed by assembly (28%) and material acquisition (18%). These results suggest that while the part itself is relatively lightweight, the environmental burden comes from how and where it's produced and shipped. The product also lacks the ability to be recycled, due to the permanent structure of the case that locks the measuring tape inside.

Material-wise, the design currently assumes the use of plastics such as PLA or ABS, chosen for their ease of prototyping and affordability. However, these materials are not biodegradable or easily recyclable in most municipal systems. The design includes multiple interfacing parts that, while effective for modularity and function, add complexity when it comes to end-of-life disassembly. It's unlikely that an average consumer would take the time to separate and properly dispose of the plastic shell, embedded metal components, and fasteners, meaning the product is more likely to end up in a landfill than a recycling facility. Given its mechanical simplicity, the product has the potential to last several years with proper use, but without a take-back program or recyclable material design, its life cycle remains linear.

While the current prototype performs well as a functional everyday tool, its environmental footprint reveals opportunities for more thoughtful material and design choices. By shifting the current materials, manufacturing methods, and packaging approach for distribution, Adapt-a-bit can evolve into a more sustainable, consumer-conscious product.

To improve sustainability, several strategic design modifications are recommended. First, switching from standard plastics like ABS or PLA to Rilsan PA11, a bio-based nylon derived from castor beans, would reduce the environmental cost of material acquisition while maintaining durability. Additionally, incorporating a clear top lid would provide visual access to the screw bits inside, improving user experience and potentially reducing unnecessary manufacturing errors. Second, redesigning for disassembly by replacing glued or rigid components with snap-fit joints or twist-lock mechanism would make teardown and recycling more feasible for consumers, decreasing the likelihood that the product ends up in a landfill.

From a production and logistics standpoint, the current system of manufacturing in China for cost efficiency, using plastic packaging, and distributing from Albany, NY comes with a high carbon footprint in the distribution phase, which accounts for 40% of the total emissions. Shifting manufacturing to the U.S. where feasible, or even nearshoring to reduce transit distances, would lessen environmental impact while improving supply chain resilience. Finally, replacing the internal metal tape measure with a plastic or aluminum alternative could lower weight and reduce material processing emissions without compromising functionality. Together, these changes would support a longer-lasting, easier-to-recycle, and more responsibly distributed product.



Final Recommendation for Product

Based on our prototype, we believe that it's best to further extend the development cycle of the product, probably for at least 8 weeks. During the first two weeks, we would focus on conducting interviews with potential users. One of our biggest challenges was getting the sizing of our case right. We would ask interviewees about what they carry with their keychains and assess how they feel about the size and weight of our prototype, given that it's meant to be carried around. We would also ask interviewees about potential design changes. We would first ask them how the material or color of a product influences how they think about a product, such as durability or practicality. We would then ask about their design preferences such as if they prefer to have multiple color options. The goal of this would be to come up with a design choice that would attract the most users.

For weeks three to five of this cycle, we would then find a way to implement these changes to improve the usability and ergonomics of our product. Based on the survey results, we would figure out the best design choice and also try to implement any size changes while keeping the case durable. We would then utilize 3D printing and other tools to create this prototype. During week six, we would then carry out tests such as a dropping test, load test, and size test and record our results. Finally, during weeks seven to eight, we would evaluate what went wrong in our tests and assess if we could make improvements.

There are three biggest aspects we would have to watch out for that could potentially extend the timeline of this cycle. First is the torque test of the screw bits; we would use the screw bit insert of the case to tighten and loosen screws. If the screw bit insert on the case becomes damaged, this would mean that we would either have to change this design or change the material, which will require further deliberation and prototyping. Second is that we decided to change the materials of the case based on our "Design for Sustainability", such changing the lid to a clear material. There is also our "Design for Manufacturing" modification, where we suggested changing the material to aluminum or tin. During the drop test and stress test, we would see how durable these materials are. Lastly, our incorporation of an automatic tape measure mechanism did not succeed initially, so based on our new "Design for Assembly" modifications and the change of the tape measure to a plastic and aluminum, we would test to see if the mechanism functions correctly after repetitive use. Based on these, the development cycle may have to be extended by a few more weeks.

Team Reflection:

Initially, our team was stuck on which idea to fully commit to during the semester. We wanted to create a multitool but didn't have a solid idea of what to incorporate. However, the creative input from our members, interviewees, and TAs helped us throughout this process. The ODP assignments, such as conducting interviews, sketching, and creating scrappy prototypes, helped us solidify our idea. It helped build a solid base foundation in the beginning of the process that we otherwise would have been lost in. Although the early stages were stressful, once we agreed on a multifunctional case, our workflow became more smooth.

One key learning on project management and team dynamics is that it's important to keep deadlines and split up the work amongst team members. When responsibilities weren't clearly defined, we found ourselves panicking as deadlines approached since tasks weren't finished on time, especially during the creation of the initial prototype in ODP 3. It was also important to set personal deadlines since the RPL had other printing jobs, which means that our waiting time was extended. Through these, we were able to learn that it's important to check-in with team members, communicate, and split up the work in order to meet our deadlines and make steady progress. Balancing other coursework sometimes made deadlines overwhelming, but in hindsight, the ODP schedule helped us maintain consistent progress.

Most of our team members started this project without having sufficient CAD and 3D printing experience. Though there were times when we were frustrated with Fusion 360, working through these challenges helped us learn critical 3D modeling skills. Additionally, our prototype is small, so we also had to carefully take into account the tolerances of 3D printers. It was stressful when the screw bits did not fit inside our 3D printed inserts. However, we were still able to continue in our process and make modifications to our prototype until we were able to get a satisfactory prototype. By the end of the project, we were able to gain a better understanding of CAD design principles and printing constraints.

Portfolios:

Emma Bourgeois (epb65) : <https://cornell-mae-ug.github.io/Emma-Bourgeois/>

Julie Nam (jsn77): <https://cornell-mae-ug.github.io/spring-2025-portfolio-JulieNam/>